C Lab Record Programs

S.No	Problem Requirement	Submission Date
11	**Swap three integers using pointers.** Write a C program that reads three integers a, b and c and rotates their values (a → b, b → c, c → a) using one temporary variable. Write a function `void swap3(int *x, int *y, int *z)` that does the swap through pointers, and call it from `main()` by passing the addresses of the three variables. Print the values before and after swapping. *Sample: Before: a=10 b=20 c=30 → After: a=30 b=10 c=20*	Nov 4, 2026
12	**Student records using a structure.** Define a `struct Student` with members `rollno` (int), `name` (char[30]), `marks[3]` (int), `total` (int) and `average` (float). Read data for 5 students into an array of structures. Write these separate functions: • `int findTotal(struct Student s)` to return the total marks • `float findAverage(int total)` to return the average • `void displayAll(struct Student s[], int n)` to print every record in a table with the columns **Roll No, Name, M1, M2, M3, Total, Average** (average to 2 decimal places). Each mark must be between 0 and 100; ask again if it isn't.	Nov 4, 2026
13	**Stack using an array (menu driven).** Write a menu-driven C program that implements a stack with a maximum size of 5 using an array and a `top` variable (starting at -1). Implement these functions: • `isFull()` returns 1 if the stack is full • `isEmpty()` returns 1 if the stack is empty • `push(int item)` inserts an element (print **"Stack Overflow"** if full) • `pop()` removes and prints the top element (print **"Stack Underflow"** if empty) • `display()` prints the elements from top to bottom Menu: *1. Push 2. Pop 3. Display 4. Exit*. Repeat the menu until the user chooses Exit, and handle invalid choices.	Nov 4, 2026
14	**Queue using an array (menu driven).** Write a menu-driven C program that implements a linear queue with a maximum size of 5 using an array with `front` and `rear` variables. Implement these functions: • `isFull()` and `isEmpty()` to check the queue's status • `enqueue(int item)` to insert at the rear (print **"Queue Overflow"** if full) • `dequeue()` to remove and print the front element (print **"Queue Underflow"** if empty) • `display()` to print the elements from front to rear Menu: *1. Enqueue 2. Dequeue 3. Display 4. Exit*. Repeat the menu until the user chooses Exit, and handle invalid choices.	Nov 4, 2026
15	**Bubble sort.** Write a C program that reads 10 integers into an array and sorts them in ascending order using bubble sort. Write a function `void bubbleSort(int arr[], int n)` for the sorting and `void printArray(int arr[], int n)` for the output. Print the array before sorting, after each pass, and after sorting. Also print the total number of swaps. *(Optional: stop early if a pass makes no swaps.)*	Nov 4, 2026